

# MonsoonIQ — Verification Report

Regime-aware post-processing of NWP rainfall · held-out period [2022, 2023] · 522 days · 27666 district-days · generated 2026-09-22 17:41 UTC

**DATA PROVENANCE: SYNTHETIC\_PHYSICALLY\_PLAUSIBLE** — Synthetic dataset generated with IMD climatology distributions, seeded physics, and documented NWP regime biases. All results are synthetic. All skill figures below are measured against this archive and are **not** evidence of operational skill.

## 1 · Scorecard (held-out period)

| Quantity                   | Raw NWP | Regime-agnostic | Regime-aware | Change vs raw         | Change vs baseline        |
|----------------------------|---------|-----------------|--------------|-----------------------|---------------------------|
| RMSE (mm/day)              | 7.352   | 3.146           | 2.786        | -4.566                | -0.360                    |
| Heavy-rain CSI (≥64.5 mm)  | 0.198   | 0.561           | 0.569        | +0.3712 (significant) | +0.0075 (not significant) |
| Heavy-rain ETS             | 0.193   | 0.555           | 0.563        | +0.369                | +0.008                    |
| Very heavy CSI (≥115.6 mm) | 0.065   | 0.160           | 0.239        | +0.174                | +0.079                    |

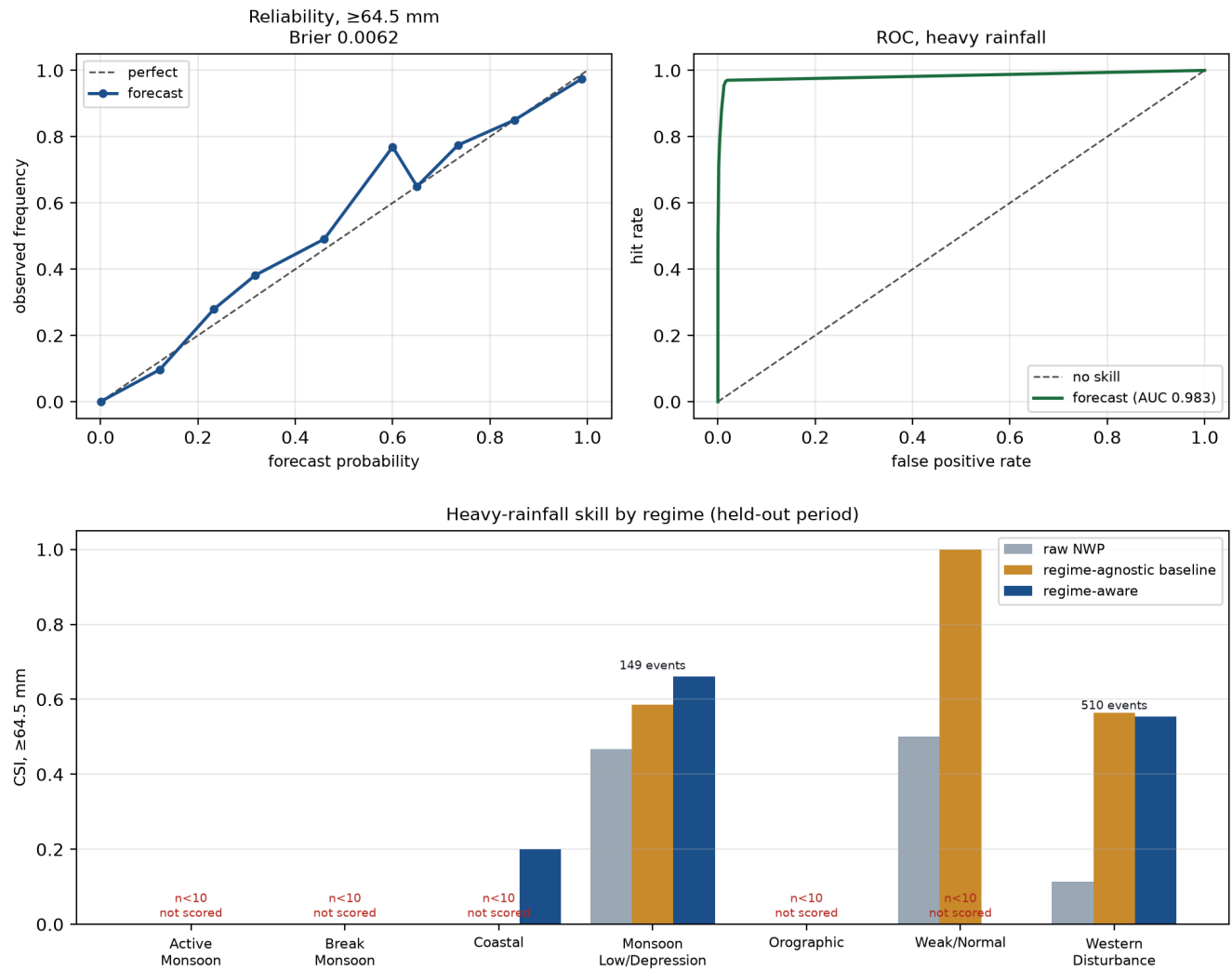
The regime-agnostic column is a gradient-boosted model fitted on the same predictors without regime information. It is the baseline that decides whether regime conditioning earns its complexity; the raw model alone does not.

## 2 · Heavy and very heavy rainfall, by regime

| Regime                 | Days | Events (≥64.5) | ≥115.6 | Reliability  | Raw CSI | Regime-agnostic CSI | Regime-aware CSI | Δ vs raw [95% CI]       |
|------------------------|------|----------------|--------|--------------|---------|---------------------|------------------|-------------------------|
| Active Monsoon         | 4134 | 5              | 0      | insufficient | 0.000   | 0.000               | 0.000            | not reportable          |
| Break Monsoon          | 2862 | 0              | 0      | insufficient | 0.000   | 0.000               | 0.000            | not reportable          |
| Coastal                | 3021 | 5              | 1      | insufficient | 0.000   | 0.000               | 0.200            | not reportable          |
| Monsoon Low/Depression | 4240 | 149            | 41     | reliable     | 0.467   | 0.586               | 0.660            | +0.194 [+0.137, +0.260] |
| Orographic             | 3233 | 1              | 0      | insufficient | 0.000   | 0.000               | 0.000            | not reportable          |
| Weak/Normal            | 4134 | 2              | 0      | insufficient | 0.500   | 1.000               | 0.000            | not reportable          |
| Western Disturbance    | 6042 | 510            | 106    | reliable     | 0.114   | 0.564               | 0.554            | +0.440 [+0.396, +0.483] |

Strata with fewer than 10 events are marked **insufficient**: their CSI is printed for completeness only and is not a scored result. Intervals are day-block bootstraps — whole calendar days are resampled, because 53 districts on one day are a single weather system.

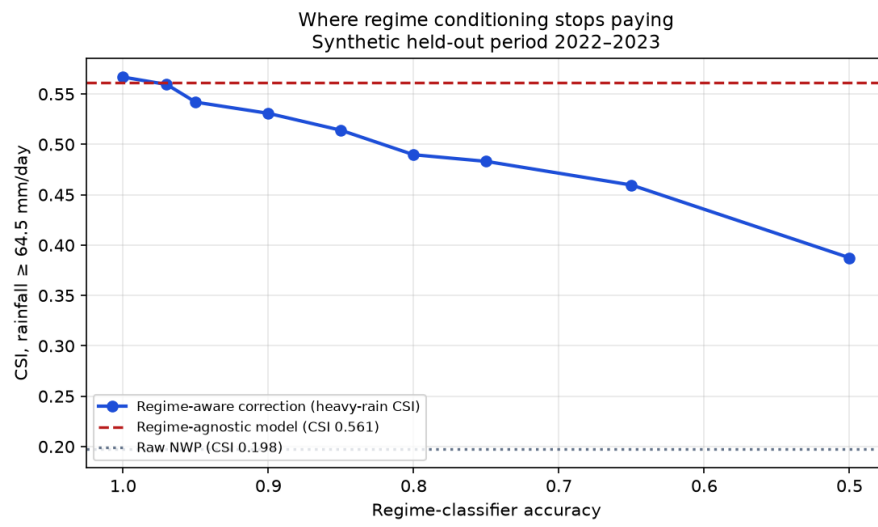
### 3 · Reliability, ROC and regime breakdown



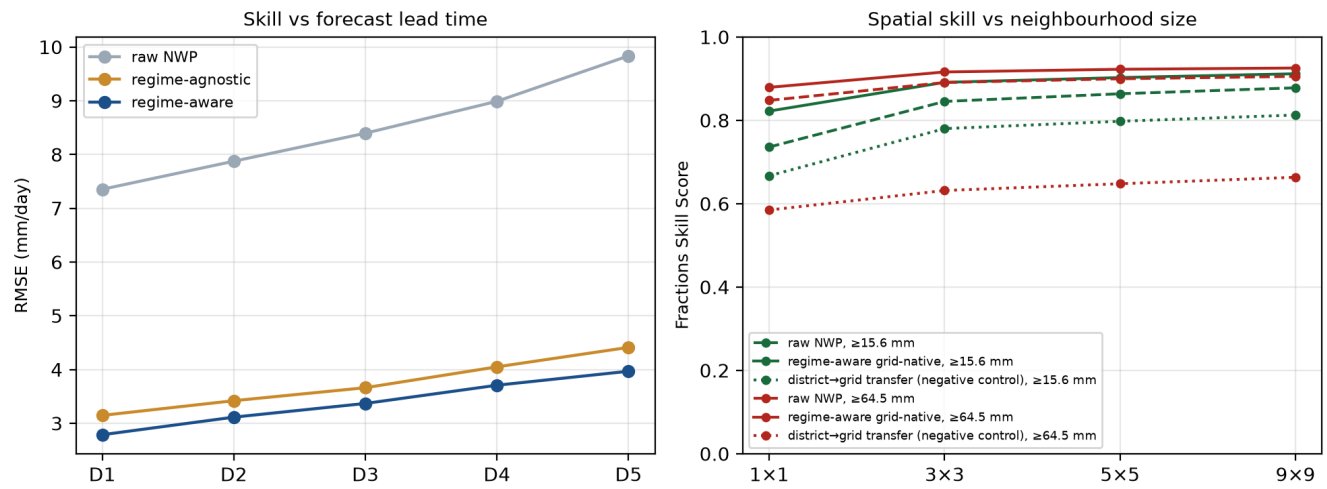
### 4 · What regime awareness is worth

| Experiment                                     | Result                                       |
|------------------------------------------------|----------------------------------------------|
| Correction alone, uniform regime weights (CSI) | 0.2111                                       |
| Correction + classifier regime weights (CSI)   | 0.5689                                       |
| Regime-agnostic model, same predictors (CSI)   | 0.5614                                       |
| Gain attributed to regime awareness            | +0.0075 [-0.0125, +0.0230] — NOT significant |
| Classifier accuracy at which the gain vanishes | 1.0                                          |

Uniform regime weights reproduce a 'correction with no regime information' system; the gap between it and the classifier blend is the actual value of regime awareness. If that gap is inside the interval, the regime layer is not yet earning its complexity and the honest fix is more training seasons or a better regime classifier - not a stronger claim.



## 5 · Lead time and spatial scale



Two grid paths are scored. `\*\_district\_transfer` extends district-scale multiplicative adjustments to cells by nearest-district assignment; `regime\_aware\_grid\_native` is a correction fitted on grid cells. Grid truth and raw grid forecast are synthetic.

| Threshold | Neighbourhood | Raw NWP | Grid-native | District→grid (control) |
|-----------|---------------|---------|-------------|-------------------------|
| ≥15.6 mm  | 1×1 cells     | 0.736   | 0.822       | 0.667                   |
| ≥15.6 mm  | 3×3 cells     | 0.845   | 0.891       | 0.780                   |
| ≥15.6 mm  | 5×5 cells     | 0.864   | 0.902       | 0.798                   |
| ≥15.6 mm  | 9×9 cells     | 0.878   | 0.912       | 0.813                   |
| ≥64.5 mm  | 1×1 cells     | 0.848   | 0.879       | 0.585                   |
| ≥64.5 mm  | 3×3 cells     | 0.891   | 0.916       | 0.631                   |
| ≥64.5 mm  | 5×5 cells     | 0.900   | 0.922       | 0.648                   |
| ≥64.5 mm  | 9×9 cells     | 0.905   | 0.925       | 0.663                   |

6 • Claim check

| Claim                                                                                                        | Verdict       | Evidence                                             |
|--------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------|
| Post-processing reduces RMSE relative to the raw model                                                       | SUPPORTED     | $\Delta$ RMSE -4.572 mm/day, 95% CI [-4.843, -4.325] |
| Post-processing improves the heavy-rainfall categorical scores relative to the raw model                     | NOT SUPPORTED | $\Delta$ CSI +0.3712, 95% CI [+0.3351, +0.4086]      |
| Regime conditioning beats a regime-agnostic model fitted to the same predictors (heavy-rainfall categorical) | NOT SUPPORTED | $\Delta$ CSI -0.0075, 95% CI [-0.0125, +0.0230]      |
| Regime conditioning beats a regime-agnostic model on RMSE                                                    | SUPPORTED     | $\Delta$ RMSE -0.365 mm/day, 95% CI [-0.501, -0.242] |

Intervals resample whole calendar days. All comparisons are against a synthetic archive, so supported claims demonstrate that the pipeline is internally consistent - not operational skill.

7 • Reproducibility

python src/train.py → python src/evaluate.py → pytest tests/ → uvicorn src.api.main:app  
Model manifest: artifacts/models/training\_manifest.json (Gradient boosting (regime-agnostic), MonsoonIQ (regime-aware), Quantile mapping (regime-agnostic), Raw NWP)  
Every figure and number in this report is regenerated from the raw archive by the commands above; no value is hand-entered.